ALICE L WALTON SCHOOL OF MEDICINE

Electrocardiogram: Fundamentals and Lead Placement (1 of 2)

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Adapted from: Electrocardiogram Part 1: Fundamentals and Lead Placement

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Diagram how action potentials translate into tracings on an electrocardiogram (ECG).
2. Explain correct lead placement for obtaining an ECG.
3. Explain how a 12-lead ECG is organized.
4. Relate the electrical events of the heart to the basic parts of an ECG.

CASE CONNECTION

To earn some extra money while in school, JD applies for a job as a nightshift ECG technician. During orientation, he is handed a packet of 10 small labeled stick-on sensors of different colors. "These are the electrodes you will use to do your job," the instructor says. "Does anyone know how they are used and how they work?" Avoiding eye contact with the instructor and hoping someone else will raise their hand, JD thinks to himself, "Maybe I should have read the orientation information packet she sent us before coming here."

How should JD have answered the instructor's questions? Consider your answers as you read, and we'll revisit JD at the end of the brick.

GO TO CONCLUSION ↓

What Is an Electrocardiogram?

The electrocardiogram (ECG) provides the link between the basic physiology of cardiac conduction (action potentials, conduction pathways) and the real world of cardiac diagnosis. Although it's tempting to simply memorize the various ECG patterns, you need to understand where all the markings on the paper come from—real cardiac myocytes depolarizing through action potentials.

In Part 1 of our discussion, we focus on the physiology—how and why the ECG works. In Part 2, we emphasize the mechanics, timing, and basic diagnosis based on the ECG. Use them together for best results!

How Do Action Gs? (LO#1)

The ECG is a composite of the electrical activity of all the millions of myocytes, summating their individual action potentials from millisecond to millisecond. Remember that myocytes transmit the action potential from cell to cell via gap junctions. So how does this morass of depolarizing cells lead to a coherent ECG that we can use to interpret conditions like myocardial infarction (MI) and arrhythmias?

Cardiac Current

When we talk about how the action potential migrates from one part of the heart to the other, we're really talking about an **electrical current**. Recall from physics that the Ohm law states:

$$\Delta V = I R$$

Where:

- ΔV is the change in voltage between two points
- I is the current
- R is the resistance in the circuit

So, for us to analyze the current that runs through the heart and is detected as an ECG, we can use a simple model of two adjacent myocardial cells, one passing the action potential to the other as part of a cardiac current, and compare their voltag us to calculate a simple two-cell ECG. These principles work remarkably well to explain the real ECG tracing.

How does the Ohm law define the voltage between two points?

Two-Cell Model

It might be hard at first to see how a normal cardiac myocyte action potential and an ECG are related (Figure 1).

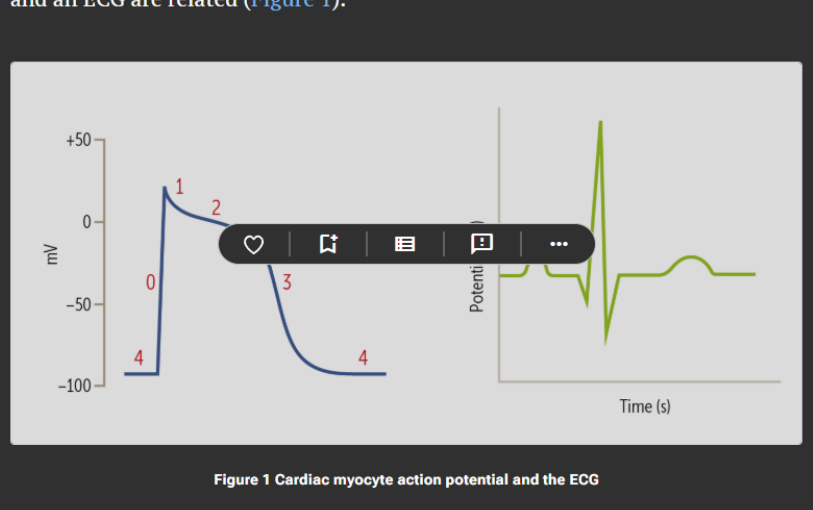


Figure 1 Cardiac myocyte action potential and the ECG

But let's do an abstract experiment that will give you an approximate idea of how this works. If we measure the voltage in two adjacent myocytes (cells A and B) as the action potential moves from one to another, we can subtract the voltage in cell B from cell A to show the change in voltage—ΔV from the Ohm equation (Figure 2).

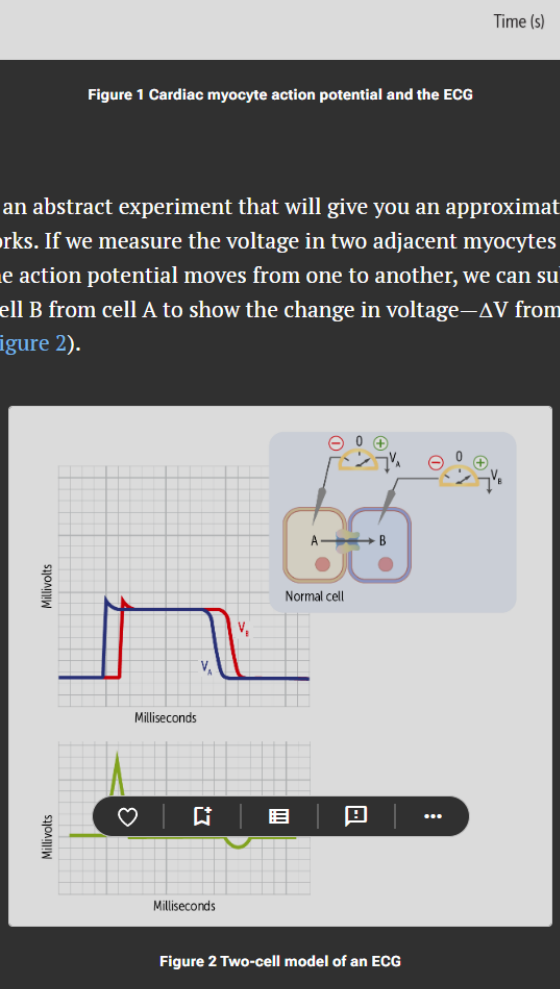


Figure 2 Two-cell model of an ECG

This is shown as subtracting the red voltage from the blue voltage, with the result in the green voltage shown at the bottom.

On the top panel in Figure 2, we see the action potentials of cells A and B side by side and offset by milliseconds, since A fires first, then passes the action potential to B through the gap junction connecting them. Their voltages are each measured with a voltmeter.

At each timepoint left to right, subtract the voltage of curve B (red) from curve A (blue) to get the voltage difference between the cells, or ΔV. That result is shown in the bottom panel of Figure 2 in green. It resembles a primitive ECG tracing, right? Let's start with the depolarization of cells A and B. Note that the upstroke of voltage in cell A is the upstroke of the "R wave" in the bottom panel, and, when cell B fires its action potential, the difference between A and B falls to baseline, which is the downstroke of the "R wave" seen in the bottom panel.

What about the repolarization of the cells (when the voltage falls)? Now, when the voltage of A falls, A – B leads to a downstroke on our mock ECG, followed by an upstroke when cell B follows suit. This creates our abstract "T wave" on the lower panel.

When you sum this ov get the real ECG. Just remember that it's ba ysics—that the difference in voltage will determine the current.

What Are the ECG Leads? (LO#2 and 3)

Before we get into what an ECG looks like, we need to know how it is generated. An electrode is a conductor placed on the patient's skin to sense the electrical activity of the heart. A lead provides a view or picture of the heart's electrical activity from different angles. The standard ECG consists of 12 leads, which means we can look at the heart's electrical activity from 12 different views. To obtain those views, we need 10 electrodes (4 on limbs, 6 on the chest).

The 12-lead ECG comprises 6 **limb leads** (I, II, III, aVR, aVL, and aVF), and 6 **chest** (or precordial) **leads** (V_1, V_2, V_3, V_4, V_5, and V_6). You will need to remember these naming conventions!

The 6 limb leads (also known as frontal leads) derive from only 4 electrodes, all in the same frontal plane. When we connect electrodes placed on the right arm, left arm, and left leg, they form the shape of a triangle, called the **Einthoven triangle** (Figure 3). It is important to note that the right leg electrode functions as an electrical ground that reduces noise and stabilizes the ECG signal but does not participate in Einthoven's triangle or in recording cardiac electrical activity.

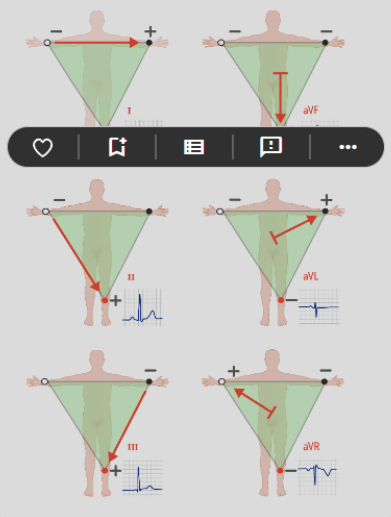
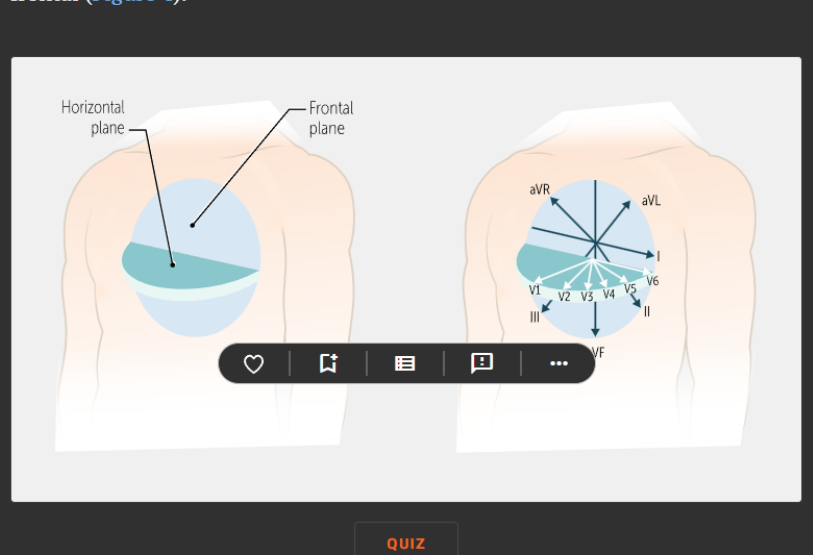


Figure 3 Einthoven triangle and ECG limb leads

The remaining precordial (chest) leads require a 3D approach to the patient. To summarize this, we can look at the heart from two planes: horizontal and frontal (Figure 4).



QUIZ

Figure 4 Heart planes and ECG chest (or precordial) leads

Each frontal plane lead represents a different viewing axis that passes through the center of Einthoven's triangle, allowing us to assess cardiac electrical activity in the frontal plane. Now, let's look at the leads in detail.

Leads I, II, and III: Bipolar Limb Leads

Leads I, II, and III are bipolar because they measure the electrical difference between a positive electrode and a negative electrode. The current travels from the negative to the positive electrode. In Figure 3:

- Lead I connects the right arm to the left arm (white dot to black dot).
  - Left arm electrode-positive electrode
  - Right are electrode-negative electrode
- Lead II connects the right arm to the left leg (white dot to red dot).
  - Left leg electrode-positive electrode
  - Right arm electrode-negative electrode
- Lead III connects the left arm to the left leg (black dot to red dot).
  - Left leg electrode-positive electrode
  - Left arm electrode-negative electrode

Leads aVR, aVL, aVF: Unipolar Limb Leads

Leads aVR, aVL, and aVF are unipolar limb leads because they record electrical activity from a single positive "exploring" electrode, referenced to a composite negative electrode formed by averaging the other two limb electrodes (Goldberger's terminal or augmented leads). This composite reference point is a mathematical construct, not a physical location in the body. The term "augmented voltage or aV" refers to amplification of the signal to increase waveform amplitude. The letters R, L, and F indicate the location of the positive electrode: right arm, left arm, and left leg (foot). In Figure 5:

- aVF: left leg electrode is the positive electrode and the composite negative electrode is the average of right arm + left arm electrodes
- aVL: left arm electrode is the positive electrode and the composite negative electrode is the average of right arm + left leg arm electrodes
- aVR: right arm electrode is the positive electrode and the composite negative electrode is the average of left leg + left arm arm electrodes

So we see, measuring the unipolar limb leads did not require additional electrodes. The six frontal plane leads are constructed using three limb electrodes—right arm, left arm, and left leg—along with a fourth electrode placed on the right leg that serves as an electrical ground. Thus, a total of four electrodes are used: right arm (RA), left arm (LA), left leg (LL), and right leg (RL). In the commonly used U.S. color-coding system, these are white, black, red, and green, respectively, with the green electrode acting as the ground. (Figure 5).

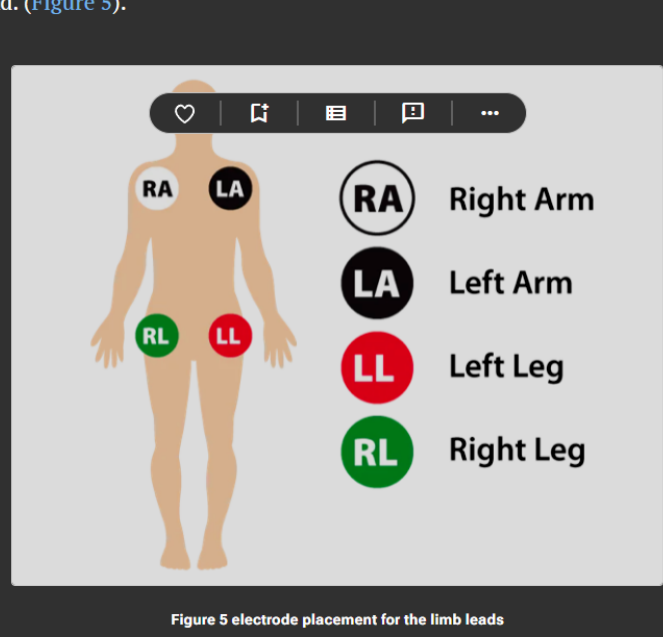


Figure 5 electrode placement for the limb leads

Mnemonic

To remember where the limb electrodes go: Smoke over fire: black (LA) over red (LL). Snow over grass: white (RA) over green (RL).

Which electrode colors correlate to which leads?

Precordial Leads: Leads V_1–V_6

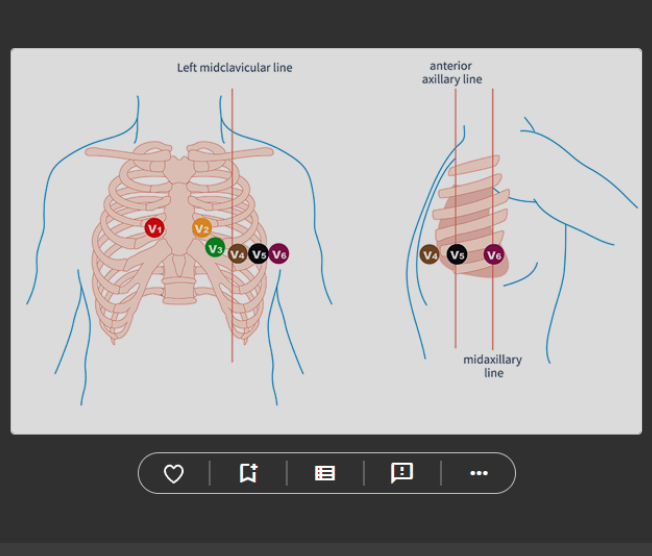
The precordial electrodes are placed across the patient's chest, beginning with V1 at the right sternal border in the fourth intercostal space and ending with V6 at the left midaxillary line (see Figure 6). Together, these six electrodes generate six precordial leads that view the heart in the horizontal (transverse) plane. The precordial leads are unipolar, meaning each uses a single positive "exploring" electrode. Wilson's central terminal is formed by averaging the voltages from all three limb electrodes.

The negative reference for the precordial leads is constructed using Wilson's central terminal, which is formed by averaging the voltages from all three limb electrodes. This differ which use Goldberger's modification that exc reference to increase signal amplitude. The precordial electrodes (ie V1-V6) all serve as the positive "exploring" electrode.

Proper electrode placement for the precordial leads are as follows (Figure 6):

- **V1: 4th intercostal space immediately to the right of the sternum**
- **V2: 4th intercostal space immediately to the left of the sternum**
- **V3: between V2 and V4**
- **V4: 5th intercostal space left midclavicular line**
- **V5: Same horizontal plane as V4, anterior axillary line**
- **V6: Same horizontal plane as V5, midaxillary line**

Proper electrode placement is important as misplacement can lead to erroneous interpretation of the 12 lead ECG. This is not just true for the precordial electrodes, but for all including the limb lead electrodes.



CLINICAL CORRELATION

The precordial leads are very important in determining the localization of myocardial ischemia.

What Does the ECG Look Like?

Now let's turn to real ECGs. The normal ECG consists of several waveforms labeled P through T. For any given ECG lead, an upward deflection occurs when a wave of depolarization moves toward the lead's positive electrode, or when a wave of repolarization moves away from it. Conversely, a downward deflection occurs when depolarization moves away from the positive electrode, or when repolarization moves toward it (Figure 6).

A biphasic waveform occurs when the net electrical vector initially moves toward the positive electrode and then reverses direction, causing the waveform to cross the baseline. This typically reflects sequential phases of depolarization and/or repolarization moving toward and away from the lead's positive electrode.

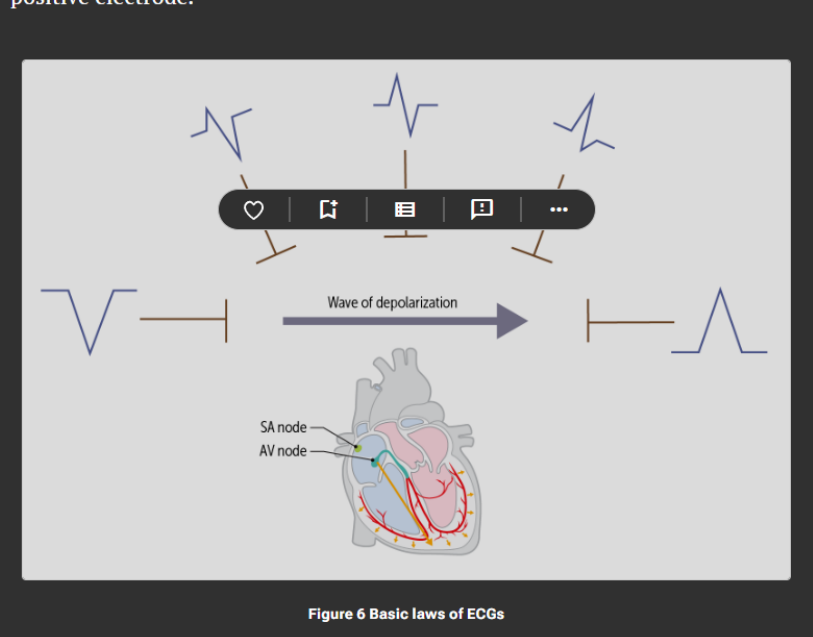


Figure 6 Basic laws of ECGs

When an impulse is traveling toward an electrode, what is the direction of the ECG wave?

Main ECG Waveforms

The main ECG waveforms are the P, QRS, and T waves. Lead II aligns well with the main direction of atrial-to-ventricular depolarization and shows the highest amplitude for these waves that we'll be studying (Figure 7, right figure, yellow arrow).

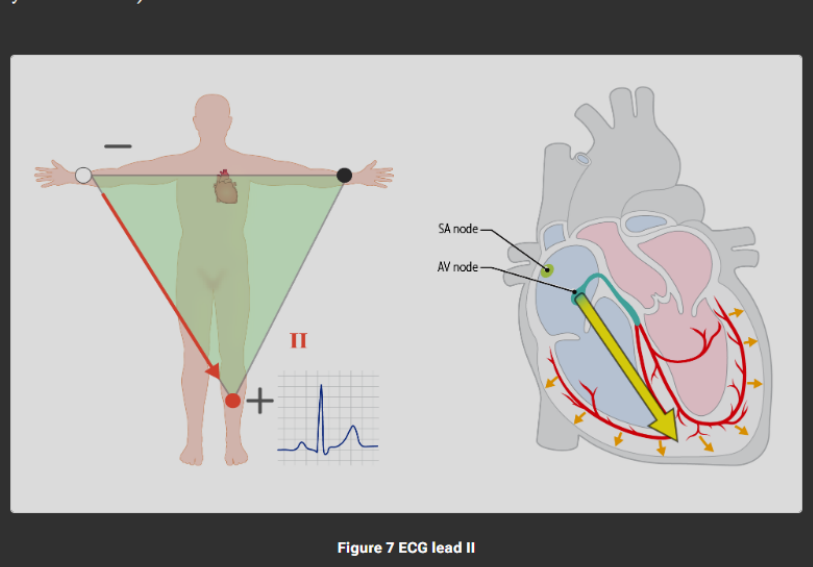


Figure 7 ECG lead II

And here is a real ECG tracing from the perspective of lead II, representing one cardiac cycle (Figure 8).

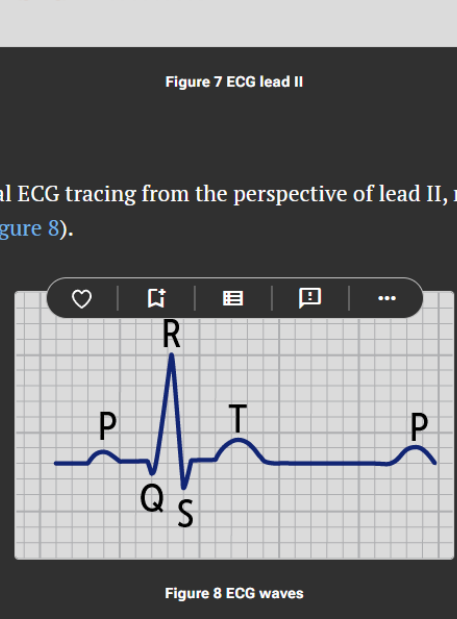


Figure 8 ECG waves

We'll go into the meaning of the various waves in detail in Part 2 of this discussion, but for now, know that the **P wave** gets things started with the atrial depolarization, followed by the **Q wave** that begins the **QRS complex** denoting ventricular depolarization. The **T wave** shows ventricular repolarization. Where is atrial repolarization? It's small and not visible, hidden within the QRS complex and its larger voltage (Figure 9).

From one P wave to the next P wave represents one cardiac cycle, <1 second of activity. This represents the electrical activity of the heart that occurs over 115,000 times per day. It must stay consistent like clockwork, or the heart stops contracting, and we lose consciousness (or die).

Correlation of the ECG With the Cardiac Myocyte Action Potential (LO#4)

Now that we've introduced the various ECG waves, let's correlate them to the underlying phenomena, which are the separate action potentials of the cells within the atrium and coordinate in time.

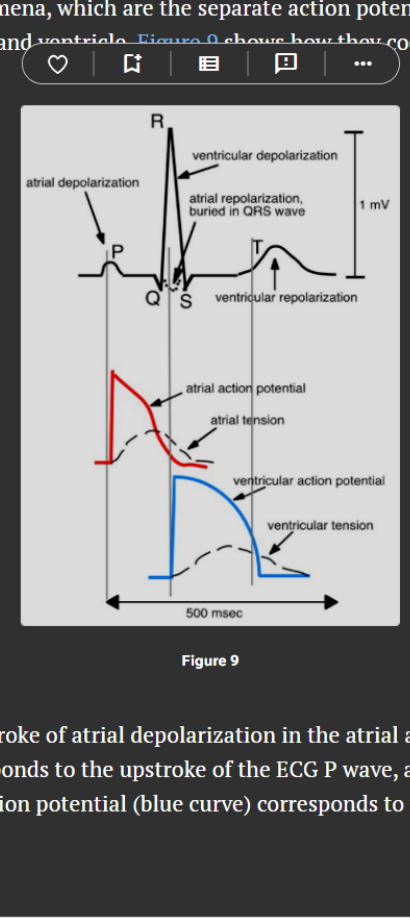


Figure 9

Note that the upstroke of atrial depolarization in the atrial action potential (red curve) corresponds to the upstroke of the ECG P wave, and the upstroke of the ventricular action potential (blue curve) corresponds to the start of the ECG QRS complex.

ACTIVE LEARNING QUESTION

Which ECG wa ization?

A. ○ P wave
B. ● QRS complex
C. ○ T wave
D. ○ PR interval
E. ○ ST segment

Explanation

The correct answer is QRS complex (B). The QRS complex represents ventricular depolarization, which is the electrical activity that triggers ventricular contraction. The P wave (A) corresponds to atrial depolarization. The T wave (C) represents ventricular repolarization. The PR interval (D) measures the time from the start of atrial depolarization to the start of ventricular depolarization. The ST segment (E) represents the period between ventricular depolarization and repolarization.

12-Lead ECG

Now let's look at an actual 12-lead ECG. The ECG is a plot of cardiac activity on a grid, with the x-axis resenting voltage (Figure 10).

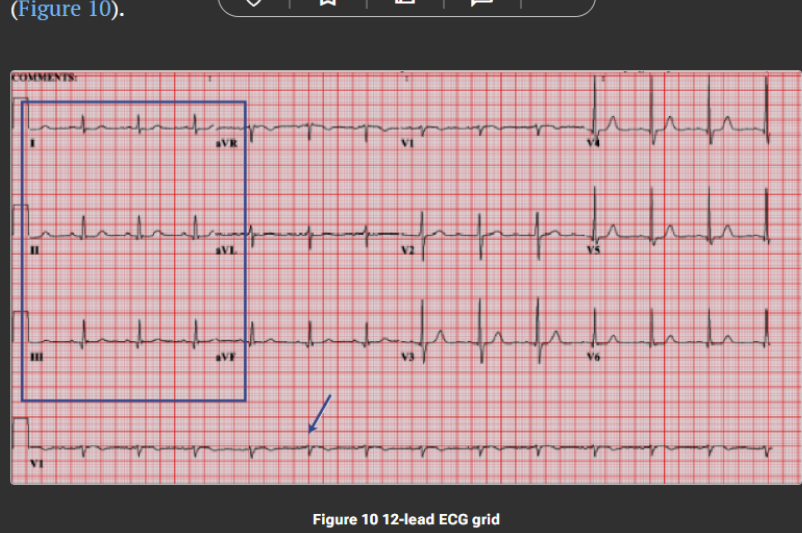


Figure 10 12-lead ECG grid

The first half of the ECG comprises of the limb leads (frontal plane) while the second half comprises of the precordial leads (horizontal plane). The leads are always organized the same way, with the first column being I, II, III; the second column being aVR, aVL, and aVF; (outlined within the blue box in Figure 10); the third column being V_1, V_2, and V_3; and the fourth column being V_4, V_5, and V_6.

At the bottom is a "rhythm strip" (arrow in Figure 10), here using lead V_1, which shows you 10 seconds of a single lead, useful for diagnosing arrhythmias. On all the other rows, eds left to right. For example, on row 1 we then V_4.

As shown in Figure 11, each of these leads show the electrical activity in the heart viewed from a different angle. Thus, it makes it important to look at all the leads.

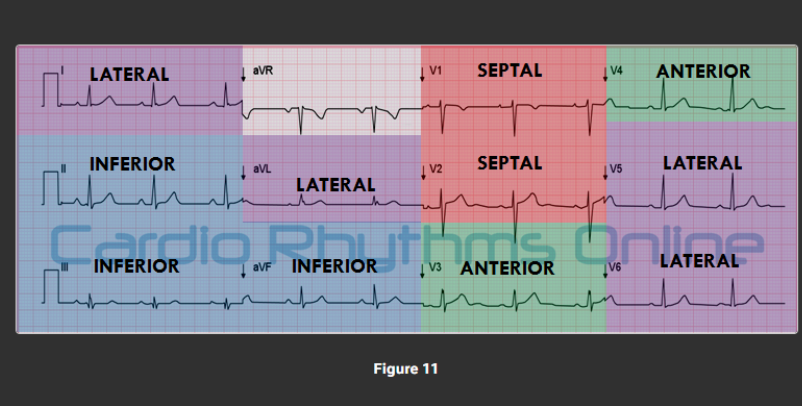


Figure 11

ECG Boxes: Timing, voltage and calibration

Timing and Voltage

On a standard 12 lead ECG, there are small boxes grouped together by larger boxes (Figure 11). Each small box is 1 mm x 1 mm which corresponds to 40 msec on the x axis and 0.1 mV on the y axis. The larger bolded boxes contain 5 x 5 small boxes, and one large box equates to 200 msec on the x axis and 0.5 mV on the y axis (Figure 12).

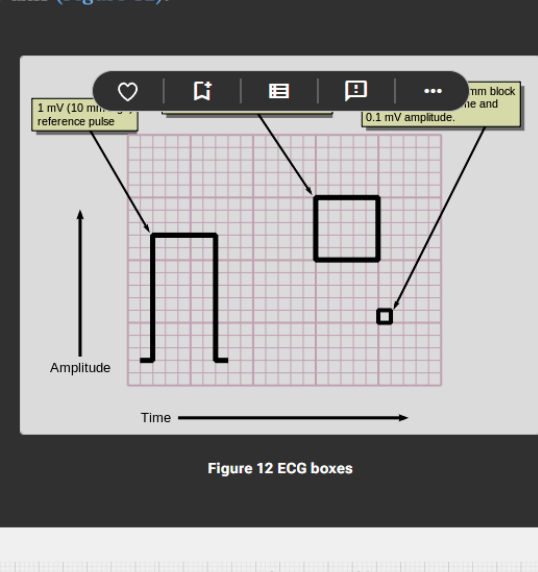
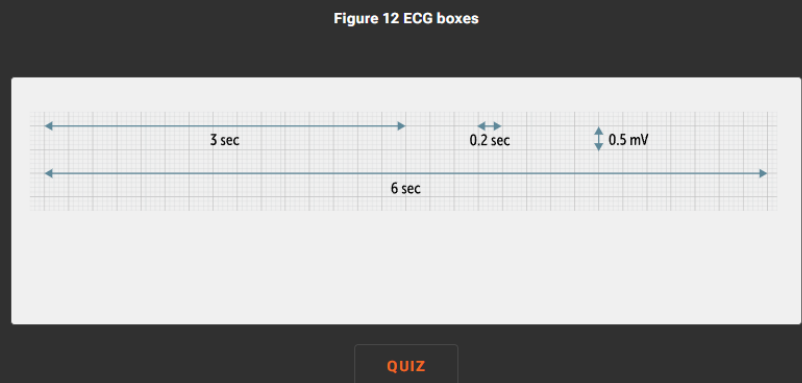


Figure 12 ECG boxes



QUIZ

Figure 13 ECG timing

Calibration

A reference pulse (or d at the beginning or the end of the ECG and is represented by a rectangular shape (Left of Figure 10). This tells us what height (mm) corresponds to 1 mV (vertical calibration) and what length (mm) corresponds to 200 msec (horizontal calibration). The standard calibration on all 12 lead ECGs is 10 mm (two large boxes) in height equals to 1 mV and 5 mm (one large box) in length equals to 200 msec. This gives us the standard sweep speed of an ECG at 25mm/s (5mm/200msec) which is often displayed at the bottom of every ECG. Of note, a standard one-page ECG recording is 250 mm in length which corresponds to a 10 second snapshot of events. As we will see later, this will be useful in estimating heart rates during irregular arrhythmias such as atrial fibrillation.

CASE CONNECTION BACK TO INTRODUCTION ↑

Thinking back to the instructor's questions, how should JD have answered?

JD now knows that the 10 electrodes (4 on the limbs and 6 on the chest) record 12 different views of the heart's electrical activity. He places lead I on the right arm, lead II on the left arm, lead III on the left leg, and a ground lead on the right leg. These leads detect electrical activity of the heart in the frontal plane. The remaining 6 electrodes are placed across the patient's chest in standardized positions, and they detect electrical activity of the heart in the horizontal plane. After JD has set up the leads on his first patient and connects them to the ECG machine, he presses the "record" button, and a 12-lead ECG and 10-second rhythm strip are printed on pink grid paper and electronically sent to the requesting physician. JD goes on to the next patient and thinks, "This is easier than I thought. I just have to make sure I pay attention to the correct lead placement each time."

Summary

- An electrocardiogram (ECG) is a representation of the electrical activity of cardiac myocytes, providing insights into cardiac conditions like myocardial infarction and arrhythmias.
- The ECG is generated by measuring voltage differences between adjacent myocytes, using principles from Ohm's law to explain the tracing.
- The 12-lead ECG uses 10 electrodes to provide 12 different views of the heart's electrical activity, comprising 6 limb leads and 6 chest/precordial leads.
- Limb leads include bipolar leads (I, II, III) and unipolar leads (aVR, aVL, aVF), while precordial leads (V1-V6) view the heart in the horizontal plane.
- Proper electrode placement is crucial for accurate ECG interpretation, with limb electrodes following color-coded conventions for placement.
- The ECG consists of waveforms labeled P through T, with upward deflections indicating depolarization moving toward the positive electrode, and the 12-lead ECG provides a 10-second snapshot of cardiac electrical activity.

Review Questions

1. How many electrodes are placed to measure a standard ECG?

A. ○ 3
B. ○ 5
C. ○ 8
D. ● 10
E. ○ 12

Hide Explanation

The correct answer is 10 (D). There are 12 commonly displayed ECG leads (I, II, III, aVL, aVR, aVF, V_1 through V_6), but these are obtained by placing 10 electrodes on the chest. This makes 3 (A), 5 (B), 8 (C), and 12 (E) incorrect.

2. What time duration is represented by a large square on an ECG grid?

A. ○ 0.04 seconds
B. ● 0.2 seconds
C. ○ 0.5 seconds
D. ○ 1 second
E. ○ 2 seconds

Hide Explanation

The correct answer is 0.2 seconds (B). Each large box on an ECG grid represents 0.2 seconds. Each small box is 0.04 seconds (A). No box value represents 0.5 seconds (C). Five large boxes equal 1 second (D), while 10 large boxes equal 2 seconds (E).

3. Which lead placement provides frontal plane views?

A. ○ Chest electrodes
B. ● Limb electrodes
C. ○ Right leg electrode
D. ○ Bypass electrodes
E. ○ Virtual electrodes

Hide Explanation

The correct answer is limb electrodes (B). Limb electrodes (B) provide a view of the heart in the frontal plane by creating the limb leads (I, II, III, aVR, aVL, aVF). Chest electrodes (A) provide horizontal plane views. The right leg electrode (C) is used as a ground. Bypass electrodes (D) are not used in standard ECGs. Virtual electrodes (E) do not exist in ECG setups.

4. Which limb is used only as a ground in ECG electrode placement?

A. ○ Left leg
B. ● Right leg
C. ○ Left arm
D. ○ Right arm
E. ○ Chest electrodes

Hide Explanation

The correct answer is right leg (B). In a standard ECG, the right leg electrode (B) is used as a ground to stabilize the signal, and does not participate in recording cardiac electrical activity. The left leg (A), left arm (C), and right arm (D) electrodes are involved in the frontal plane views. The chest electrodes (E) do not participate in the limb lead setup.

5. What the correct placement for lead V1?

A. ○ Left midclavicular line, 5th intercostal space
B. ● Right sternal border, 4th intercostal space
C. ○ Left sternal border, 4th intercostal space
D. ○ Midaxillary line, same level as V4

Hide Explanation

The correct answer is lead V1 is placed at the right sternal border in the 4th intercostal space (B), which optimally records electrical activity from the interventricular septum and right ventricle, making it essential for accurate assessment of septal depolarization and conduction abnormalities. The left midclavicular line at the 5th intercostal space (A) describes the placement of V4, not V1, which is positioned at the right sternal border. The left sternal border at the 4th intercostal space (C) corresponds to V2, whereas V1 must be placed on the right side of the sternum at the same intercostal level. The midaxillary line at the same horizontal level as V4 (D) describes the placement of V6, not V1, and placing V1 laterally would significantly alter septal lead interpretation.

6. Which ECG wave corresponds to ventricular depolarization?

A. ○ P wave
B. ● QRS complex
C. ○ T wave
D. ○ PR interval
E. ○ ST segment

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The correct answer is QRS complex (B). The QRS complex represents ventricular depolarization, which is the electrical activity that triggers ventricular contraction. The P wave (A) corresponds to atrial depolarization. The T wave (C) represents ventricular repolarization. The PR interval (D) measures the time from the start of atrial depolarization to the start of ventricular depolarization. The ST segment (E) represents the period between ventricular depolarization and repolarization.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

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02 ☑ ~~Explain correct lead placement for obtaining an ECG.~~
03 ☑ ~~Explain how a 12-lead ECG is organized.~~
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Objective Confidence: 4/4

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